

Math 199 (Math 231 Section AD2) Syllabus Fall 2022

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Office Hours: Wednesday, 4-5pm, Altgeld Hall, Room 173
Friday, 2-3pm, English Building, Room 304.

Website: <https://yigalkamel.web.illinois.edu/M199Fa22Page.html>

Meeting Times: Tuesday and Thursday, 9am-10:50am

Meeting Location: Altgeld Hall, Room 159

Welcome to Calculus II! In this class we will delve further into the realm of infinite mathematics.

If you are in this class, hopefully you have some taste (from Calculus I, or elsewhere) of how we deal with “infinite constructions” in mathematics (using limits), and maybe you’ve seen how useful these techniques can be to solve problems! The first examples of such concepts to do with infinity that you’ve seen were limits (of functions at a point), continuity, derivatives, and integrals. In this class, we will learn about more ways to compute integrals, more ways to *use* integrals to compute other things we might be interested in, and then we’ll move on to *new kinds* of infinite constructions called (infinite) **sequences** and **series**.

A sequence is just a(n) (infinite) list of numbers. The study of sequences enters the world of calculus when the sequences in question behave in a special way as you move further and further along in the sequence. Particularly, if the numbers in the sequence get closer and closer to a specific number (in a precise sense), we might say the sequence **converges** to this number. This special number, in turn, is called a **limit**! (You might notice that this use of the word limit is *different* from the kinds of limits you’ve seen in Calculus I, even though the idea is similar.)

While these infinite sequences may sound interesting to you in their own right, we are really going to be interested in particular kinds of sequences called **series**. A series is an attempt to make sense of an “infinite sum” using the machinery of calculus. We do this by imagining the sequence of finite sums we get by adding another term at each step. You may wonder what it is about series that we should care about? Well, when we start to consider series of *functions* instead of just numbers, we’ll find new ways to represent many of the familiar differentiable functions as series, which will help us learn new things about them.

What is Math 199? What is a Merit discussion section?

The 4 hours that we will spend together each week serve two separate courses.

For roughly 2 of these hours, our class is discussion section AD2 of Math 231 (Calculus 2), and I am your TA. This means that for that time, we will be following along with the worksheets assigned for Math 231, and your performance on these worksheets will be relevant to your Math 231 grade, as per the guidelines of that course. I do not make the content for this part of the course, and I do not assign your grades.

For the remaining 2 hours, our class is Math 199, and I am your instructor. This is the time that the word “merit” in the name of this discussion section offers us to explore the material in Calculus II in

a slightly different way. During this time, we will gain a deeper understanding of the material by

1. solving difficult and interesting problems by working together in groups, and
2. thinking critically about the motivation for and underlying purpose of the concepts we will study.

Part of the philosophy of the merit program is that collaborative “stuckness” is a necessary component of learning mathematics. By this I mean: struggling with a problem or concept is inevitable, and you’ll benefit greatly by working through such difficulties with your fellow students. This will involve sharing your ideas, as well as listening to your group mates’ ideas. The more paths towards a solution that you encounter, the better you’ll know the landscape. In this spirit, I will often not give you the answers/solutions to the problems we will work on, but I *will* do my best to help guide you through your collective journey into this new part of mathematics.

Grade breakdown:

This is for your Math 199 grade. Your Math 231 grade is unrelated.

Attendance: 20%

Participation: 50%

Preparation: 30%

Attendance: You will be graded on attendance.

I will drop 2 absences, no questions asked.

An excused absence will not count as a drop, but will require either verified documentation, or approval from me in advance.

Lateness: 20 minutes late = full absence and maximum 50% for that day’s preparation/participation scores.
5 minutes late = half an absence.

Participation: Your score for this part of the grade will be based on your participation with (and submission of) any activities we do *in class*. This includes worksheets, or other activities that we may do. You can raise this part of your grade by demonstrating your participation to me in other ways (e.g. coming to office hours, and being engaged with the material).

I will drop 2 participation assignment submissions.

For any additional excused absences, talk to me about making up the participation points.

Preparation: Your score for this part of the grade will be based on the work you do *before class* to prepare for class. This will include short assignments I will ask you to do (and hand in) to help stimulate class discussions about important topics. As above, you can raise this grade above your raw score by demonstrating your preparation in other ways.

I will drop 2 preparation assignment submissions.

For any additional excused absences, talk to me about making up the preparation points.

I hope you are all excited for an exciting semester of math and this great opportunity you have to engage with the material in a unique way. I am excited to get to know all of you and help you along as you progress through the course!